

Answer:

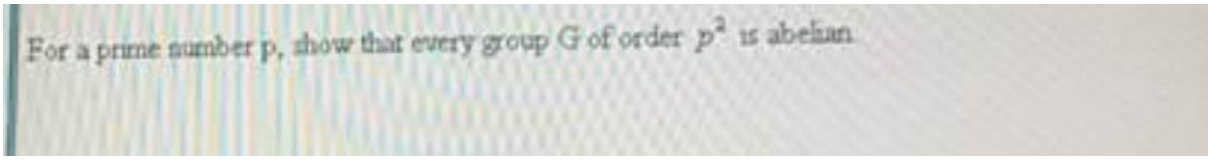
Solution

We need to show that $y(s+t)=y(s)+y(t)$ for $s, t \in \mathbb{Z}$. Using the division algorithm, we let

$s=q_1n+r_1$ (1) and

$t=q_2n+r_2$ (2) where $0 \le r_i < n$ for $i=1, 2$.

If $r_1+r_2=q_3n+r_3$ (3) for $0 \le r_3 < n$ then adding Eqs. (1) and (2) we see that $s + t = (q_1 + q_2 + q_3)n + r_3$, so that $y(s+t)=r_3$. From Eqs. (1) and (2) we see that $y(s) =r_1$ and $y(t)=r_2$. Equation (3) shows that the sum r_1+r_2 in $\mathbb{Z}_n$ is equal to r_3 also.



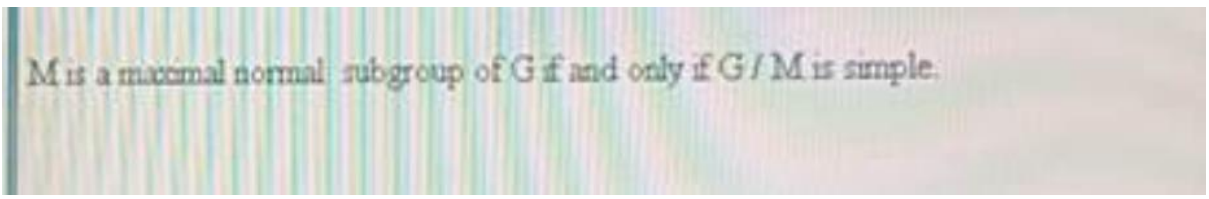
Answer:

Proof

If G is not cyclic, then every element except e must be of order p.

Let a be such an element. Then the cyclic subgroup $\langle a \rangle$ of order p does not exhaust G.

Also let $b \in G$ with $b \notin \langle a \rangle$. Then $\langle a \rangle \cap \langle b \rangle = \{e\}$, since an element c in $\langle a \rangle \cap \langle b \rangle$ with $c \neq e$ would generate both $\langle a \rangle$ and $\langle b \rangle$, giving $\langle a \rangle = \langle b \rangle$, contrary to construction.



Answer:

Proof

Let M be a maximal normal subgroup of G. Consider the canonical homomorphism

$y: G \to G/M$. Now y^{-1} of any nontrivial proper normal subgroup of G/M is a proper normal subgroup of G properly containing M. But M is maximal, so this can not happen. Thus G/M is simple.

Give an example to show that it is false that if d divides the order of G, then there must exist a subgroup H of G having order d.

Answer:

Example

We show that A_4, which has order 12, contains no subgroup of order 6.

Suppose that H were a subgroup of A_4 having order 6.

If N is a normal subgroup of G and H is also normal in G, then show that HN is normal in G.

Answer (Please click here to Add Answer)

100%

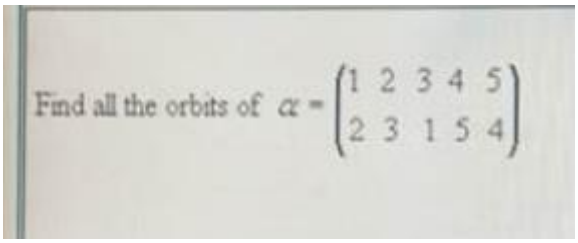
Answer:

Lemma

If N is a normal subgroup of G, and if H is any subgroup of G, then

H V N=HN=NH.

Furthermore, if H is also normal in G, then HN is normal in G.

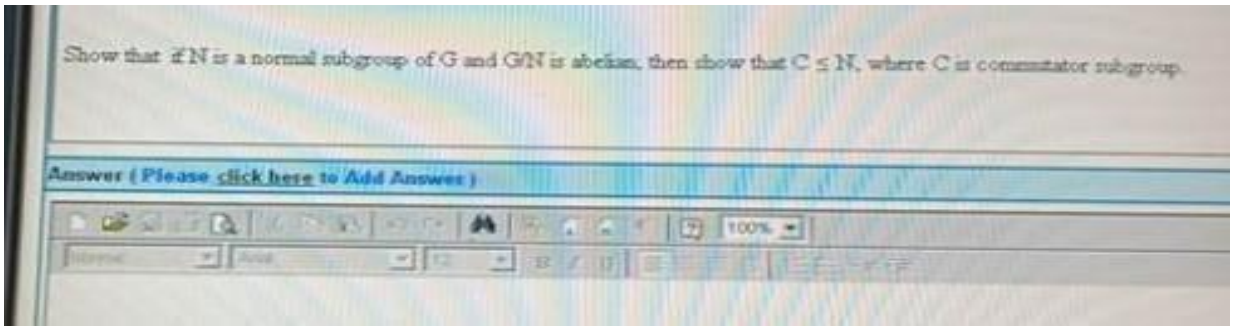


Answer:

- Look at what happens to elements as a permutation is applied.
- $$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

$\alpha(1)=2, \alpha^2(1)=3, \alpha^3(1)=1$ {1,2,3}

$\alpha(4)=5, \alpha^2(4)=4$ {4,5}



ANSWER:

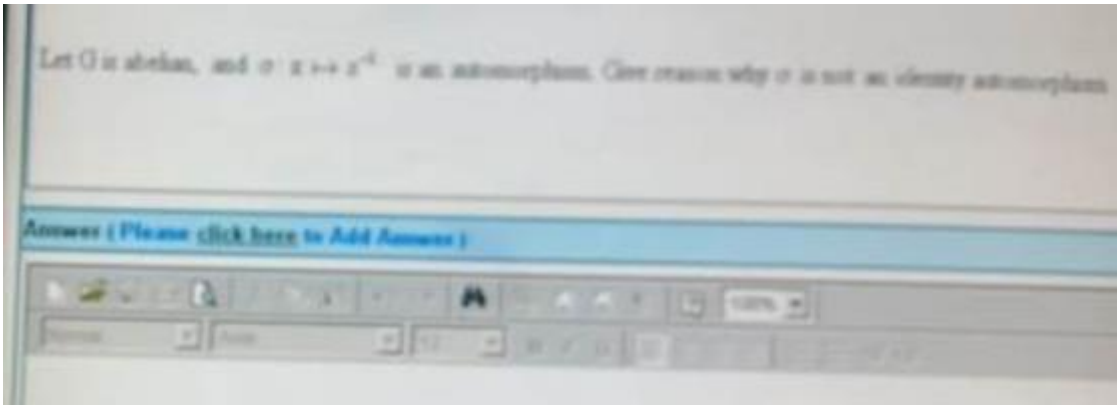
Proof

If N is a normal subgroup of G and G/N is abelian, then

$(a^{-1}N)(b^{-1}N)=(b^{-1}N)(a^{-1}N)$;
that is, $aba^{-1}b^{-1}N=N$,

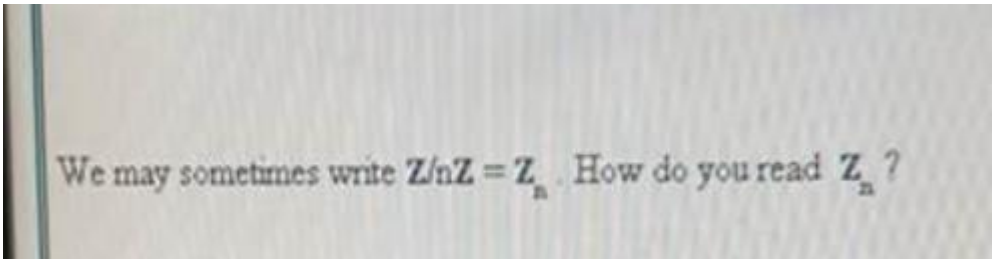
so $aba^{-1}b^{-1} \in N$, and

$C \leq N$.



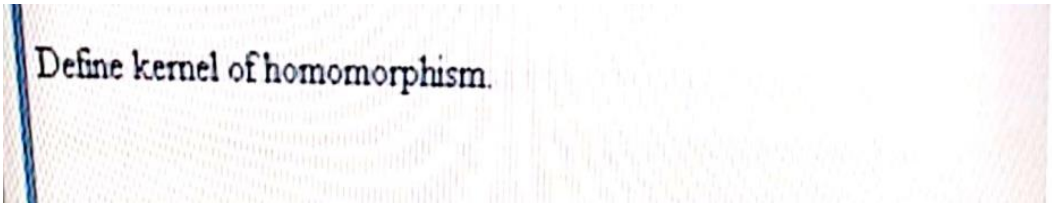
Answer:

When G is abelian, then $\sigma: x \mapsto x^{-1}$ is an automorphism, and, clearly, σ is not an identity automorphism.
When G is not abelian, there exists a nontrivial inner automorphism.



Answer:

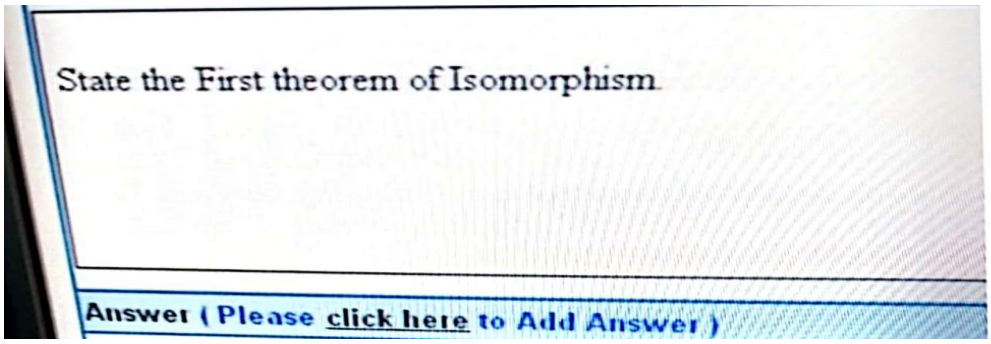
Theorem, we see that the factor group $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. The cosets of $n\mathbb{Z}$ are the residue classes modulo n.



Answer:

Definition

Let h: G→G' be a homomorphism of groups. The subgroup $h^{-1}[\{e'\}]=\{x\in G \mid h(x)=e'\}$ is the **kernel** of h, denoted by Ker(h).



Answer:

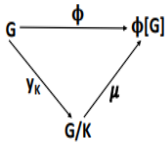
Theorem

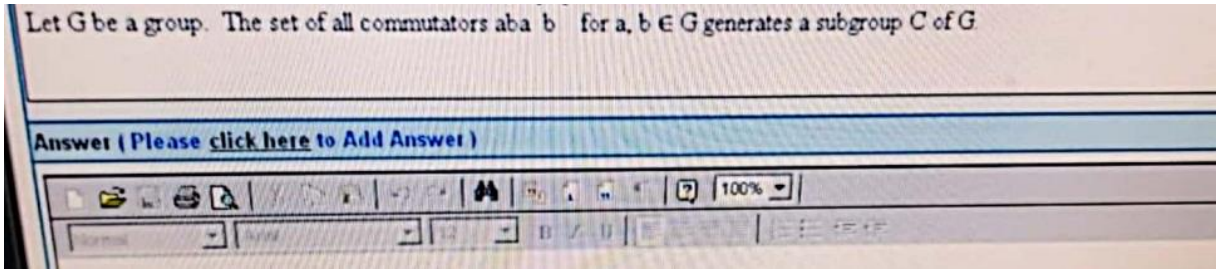
Let ϕ: G→G' be a homomorphism with kernel K, and let

γ_K: G → G/K be the canonical homomorphism. There is a unique isomorphism

μ: G/K→ ϕ[G] such that $\phi(x) = \mu(\gamma_K(x))$ for each x∈G.

The first isomorphism theorem is diagrammed in Figure below.





Answer:

The Commutator Subgroup

Proof

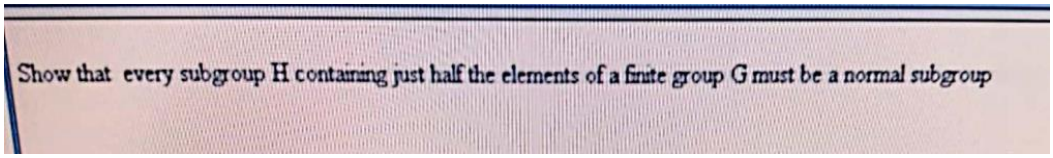
Let a, b ∈ G. Then,

$(aba^{-1}b^{-1})(aba^{-1}b^{-1})^{-1}$

$=aba^{-1}b^{-1}bab^{-1}a^{-1}$

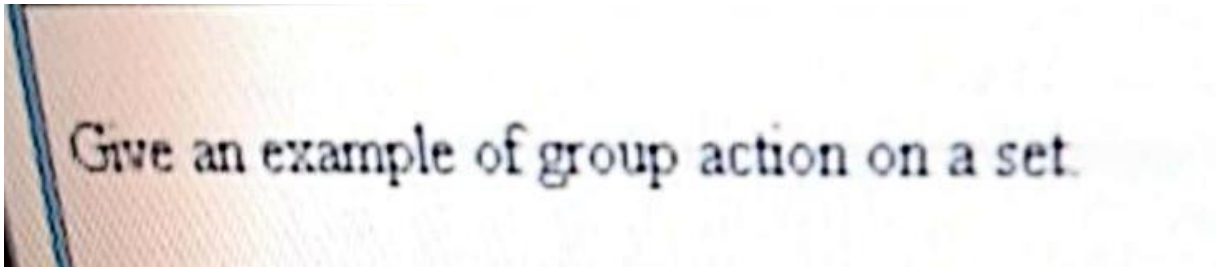
$=e \in C$

since $e = eee^{-1}e^{-1}$ is a commutator.



Answer:

Note that every subgroup H containing just half the elements of a finite group G must be a normal subgroup, since for each element a in G but not in H, both the left coset aH and the right coset Ha must consist of all elements in G that are not in H.

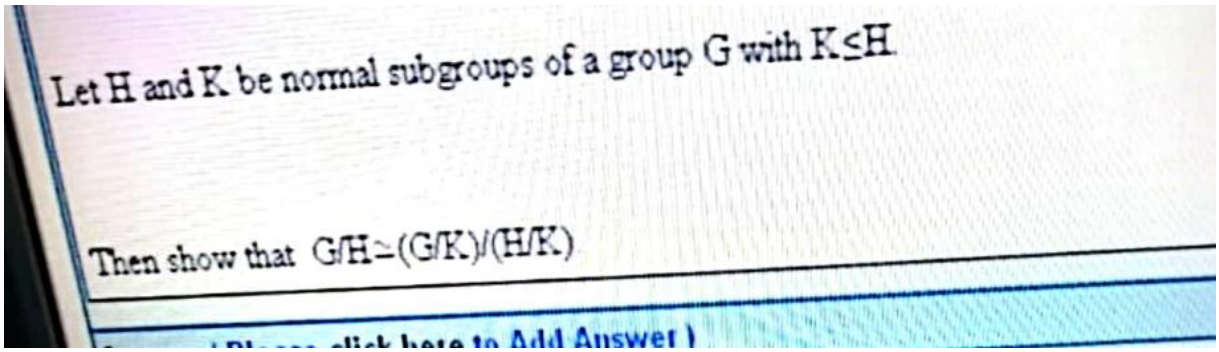


Answer:

Example

Let X be any set, and let H be a subgroup of the group S_x of all permutations of X.

Then X is an H -set, where the action of $\sigma \in H$ on X is its action as an element of S_x, so that $\sigma x = \sigma(x)$ for all $x \in X$.



Answer:

Proof

Let $\phi: G \to (G/K)/(H/K)$ be given by $\phi(a) = (aK)(H/K)$ for $a \in G$.

Clearly ϕ is onto $(G/K)/(H/K)$, and for $a, b \in G$,

$\phi(ab) = [(ab)K](H/K)$

$= [(aK)(bK)](H/K)$

$= [(aK)(H/K)][(bK)(H/K)] = \phi(a)\,\phi(b)$,

so ϕ is a homomorphism.

Show that the homomorphism $h : G \to G'$ is injective if and only if $Ker\ h=(e)$.

Answer:

Proof

Suppose h is injective,

and let $x \in Ker\ h$.

Then $h(x)=e'=h(e)$.

Hence $x=e$.